

Panel Data Models

Dr. Randall R. Rojas

Contents

1	Introduction	1
2	A Microeconomic Panel	1
3	A Pooled Model	2
4	The Fixed Effects Model	7
4.1	The Least Squares Dummy Variable Estimator	7
4.2	The Fixed Effects (Within) Estimator	11
5	The Random Effects Model	13
5.1	Several Correlations to be Considered:	14
6	Comparing Fixed and Random Effects Estimators	16
7	The Hausman-Taylor Estimator	19
8	Sets of Regression Equations	21

1 Introduction

- A panel of data consists of a group of cross-sectional units (people, households, firms, states, countries) who are observed over time
- Denote the number of cross-sectional units (individuals) by N
- Denote the number of time periods in which we observe them as T
- Different ways of describing panel data sets:
 - Long and narrow:
“Long” describes the time dimension and “narrow” implies a relatively small number of cross sectional units
 - Short and wide
There are many individuals observed over a relatively short period of time
 - Long and wide
Both N and T are relatively large
- It is possible to have data that combines cross-sectional and time-series data which do not constitute a panel

- We may collect a sample of data on individuals from a population at several points in time, but the individuals are not the same in each time period

2 A Microeconomic Panel

- In microeconomic panels, the individuals are not always interviewed the same number of times, leading to an *unbalanced panel* in which the number of time series observations is different across individuals
- In a *balanced panel*, each individual has the same number of observations

3 A Pooled Model

- A *pooled model* is one where the data on different individuals are simply pooled together with no provision for individual differences that might lead to different coefficients

$$y_{it} = \beta_1 + \beta_2 x_{2it} + \beta_3 x_{3it} + e_{it} \quad (1)$$

- Note that the coefficients $(\beta_1, \beta_2, \beta_3)$ do not have i or t subscripts
- The least squares estimator, when applied to a pooled model, is referred to as *pooled least squares*
- The data for different individuals are pooled together, and the equation is estimated using least squares
- It is useful to write explicitly the error assumptions required for pooled least squares to be consistent and for the t and F statistics to be valid when computed using the usual least squares variance estimates and standard errors

$$E(e_{it}) = 0$$

$$Var(e_{it}) = E(e_{it}^2) = \sigma_e^2$$

$$Cov(e_{it}, e_{is}) = E(e_{it}e_{is}) = 0, \quad \text{for } i \neq j, \text{ or } t \neq s$$

$$Cov(e_{it}, x_{2it}) = Cov(e_{it}, x_{3it}) = 0$$

- Applying pooled least squares in a way that ignores the panel nature of the data is restrictive in a number of ways
- The first unrealistic assumption that we consider is the lack of correlation between errors corresponding to the same individual
- To relax the assumption of zero error correlation over time for the same individual, we write:

$$Cov(e_{it}, e_{is}) = \psi_{ts} \quad (2)$$

- This also relaxes the assumption of homoskedasticity:

$$Cov(e_{it}, e_{it}) = Var(e_{it}) = \psi_{tt} \quad (3)$$

- We continue to assume that the errors for different individuals are uncorrelated:

$$Cov(e_{it}, e_{js}) = E(e_{it}e_{js}) = 0, \quad \text{for } i \neq j$$

- What are the consequences of using pooled least squares in the presence of the heteroskedasticity and correlation?

- The least squares estimator is still consistent
- Its standard errors are incorrect
 - This implies that hypothesis tests and interval estimates based on these standard errors will be invalid
 - Typically, the standard errors will be too small, overstating the reliability of the least squares estimator
- Standard errors that are valid for the pooled least squares estimator under the assumption in Eq. 2 can be computed
- Various names are:
 - Panel-robust standard errors
 - Cluster-robust standard errors

The time series observations on individuals are the clusters

Pooled Model Summary

Take the entire data set and simply run a regression (using OLS) by pooling all observations together.

$$Y_{it} = \alpha + \beta X_{it} + e_{it}$$

When we run a pooled model, we impose the following assumptions about the data generating process:

1. There exists a constant relationship between **X** and **Y** across time (i.e., β does not change over time)
2. There exists a constant relationship between **X** and **Y** across sub-groups (i.e., β does not change over the different units in the data set)

R Example

```
rm(list=ls())
library(plm) # Linear Models for Panel Data
library(AER)
library(POE5Rdata)

# Read in the data
data("nls_panel", package="POE5Rdata")
head(nls_panel)
```

```
##   id year   lwage hours age educ collgrad msp nev_mar not_smsa c_city south
## 1  1  82 1.808289   38  30  12         0  1         0         0         1     0
## 2  1  83 1.863417   38  31  12         0  1         0         0         1     0
## 3  1  85 1.789367   38  33  12         0  0         0         0         1     0
## 4  1  87 1.846530   40  35  12         0  0         0         0         1     0
## 5  1  88 1.856449   40  37  12         0  0         0         0         1     0
## 6  2  82 1.280933   48  36  17         1  1         0         0         0     0
##  black union    exper    exper2   tenure   tenure2
```

```
## 1      1      1  7.666667  58.77777 7.666667 58.777770
## 2      1      1  8.583333  73.67361 8.583333 73.673610
## 3      1      1 10.179490 103.62200 1.833333  3.361111
## 4      1      1 12.179490 148.33990 3.750000 14.062500
## 5      1      1 13.621790 185.55330 5.250000 27.562500
## 6      0      0  7.576923  57.40976 2.416667  5.840278
```

```
dim(nls_panel)
```

```
## [1] 3580    18
```

```
# Convert the data to a panel data structure where i = id, and t = year
```

```
nls.pd <- pdata.frame(nls_panel, index=c("id", "year"))
head(nls.pd)
```

```
##      id year    lwage hours age educ collgrad msp nev_mar not_smsa c_city south
## 1-82  1   82 1.808289    38  30   12         0   1         0         0         1     0
## 1-83  1   83 1.863417    38  31   12         0   1         0         0         1     0
## 1-85  1   85 1.789367    38  33   12         0   0         0         0         1     0
## 1-87  1   87 1.846530    40  35   12         0   0         0         0         1     0
## 1-88  1   88 1.856449    40  37   12         0   0         0         0         1     0
## 2-82  2   82 1.280933    48  36   17         1   1         0         0         0     0
##      black union    exper    exper2    tenure    tenure2
## 1-82      1     1  7.666667  58.77777 7.666667 58.777770
## 1-83      1     1  8.583333  73.67361 8.583333 73.673610
## 1-85      1     1 10.179490 103.62200 1.833333  3.361111
## 1-87      1     1 12.179490 148.33990 3.750000 14.062500
## 1-88      1     1 13.621790 185.55330 5.250000 27.562500
## 2-82      0     0  7.576923  57.40976 2.416667  5.840278
```

```
dim(nls.pd)
```

```
## [1] 3580    18
```

```
# Estimate the Pooled OLS
```

```
mreg.pooled <- plm(lwage~educ+exper+I(exper^2)+
  tenure+I(tenure^2)+black+south+union,
  model="pooling", data=nls.pd)
summary(mreg.pooled)
```

```
## Pooling Model
```

```
##
```

```
## Call:
```

```
## plm(formula = lwage ~ educ + exper + I(exper^2) + tenure + I(tenure^2) +
##      black + south + union, data = nls.pd, model = "pooling")
```

```
##
```

```
## Balanced Panel: n = 716, T = 5, N = 3580
```

```
##
```

```
## Residuals:
```

```
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -1.6956994 -0.2331132 -0.0043786  0.2151676  2.5816751
```

```
##
```

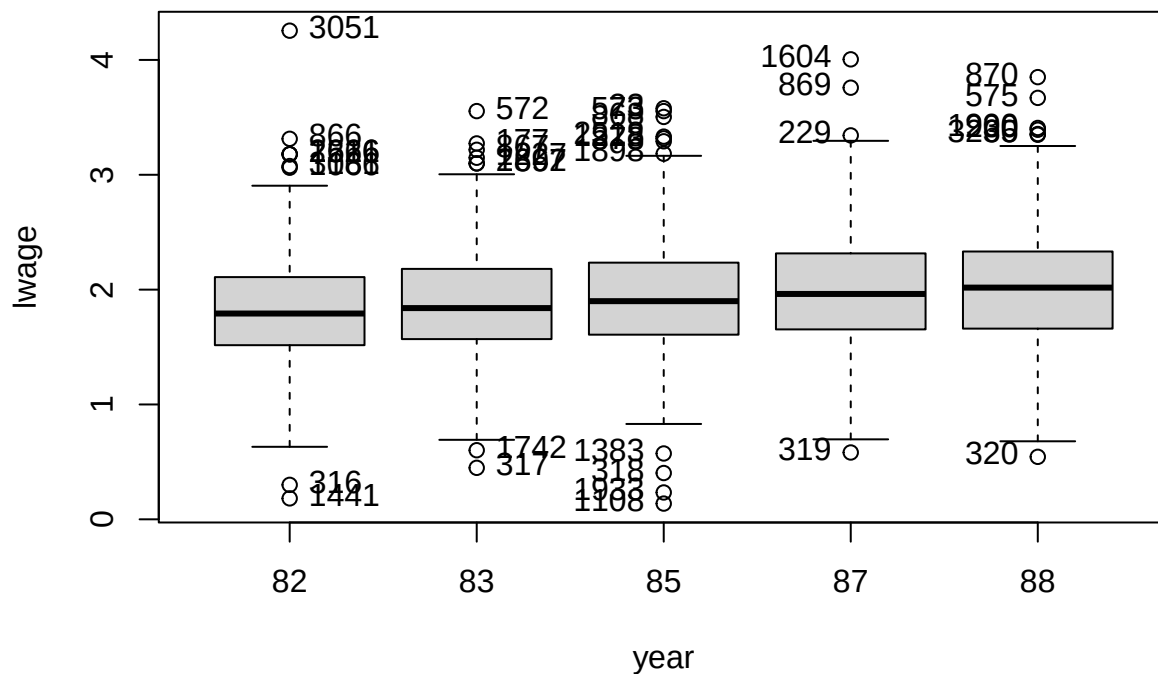
```
## Coefficients:
```

```
##              Estimate Std. Error t-value Pr(>|t|)
## (Intercept)  0.47660008  0.05615585   8.4871 < 2.2e-16 ***
## educ         0.07144879  0.00268939 26.5669 < 2.2e-16 ***
## exper        0.05568504  0.00860716  6.4696 1.116e-10 ***
```

```
## I(exper^2) -0.00114754 0.00036129 -3.1762 0.0015046 **
## tenure 0.01496002 0.00440728 3.3944 0.0006953 ***
## I(tenure^2) -0.00048604 0.00025770 -1.8860 0.0593697 .
## black -0.11671387 0.01571590 -7.4265 1.387e-13 ***
## south -0.10600256 0.01420083 -7.4645 1.045e-13 ***
## union 0.13224321 0.01496161 8.8388 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares: 772.56
## Residual Sum of Squares: 521.03
## R-Squared: 0.32559
## Adj. R-Squared: 0.32408
## F-statistic: 215.496 on 8 and 3571 DF, p-value: < 2.22e-16
# Estimate the Pooled OLS w/ Cluster-robust standard errors
coeftest(mreg.pooled, vcov=vcovHC(mreg.pooled,
type="HC0",cluster="group"))
```

```
##
## t test of coefficients:
##
## Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.47660008 0.08440936 5.6463 1.767e-08 ***
## educ 0.07144879 0.00548952 13.0155 < 2.2e-16 ***
## exper 0.05568504 0.01128960 4.9324 8.492e-07 ***
## I(exper^2) -0.00114754 0.00049158 -2.3344 0.01963 *
## tenure 0.01496002 0.00711024 2.1040 0.03545 *
## I(tenure^2) -0.00048604 0.00040948 -1.1870 0.23532
## black -0.11671387 0.02808308 -4.1560 3.314e-05 ***
## south -0.10600256 0.02701241 -3.9242 8.865e-05 ***
## union 0.13224321 0.02702550 4.8933 1.036e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

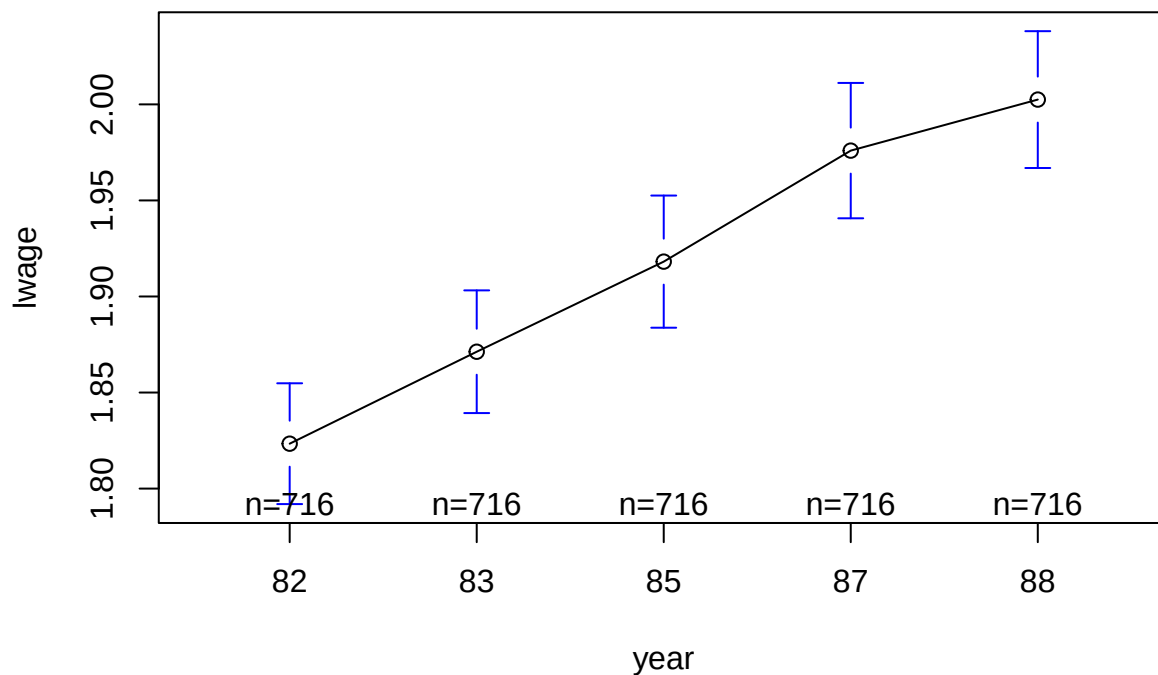
```
# We can start be exploring heterogeneity across time (year) and individuals (id)
library(gplots)
library(car)
# Heterogeneity across time:
scatterplot(lwage~year|id, data=nls.pd)
```



```
## [1] "316" "1441" "866" "1066" "1226" "2516" "2661" "3051" "3181" "317"
## [11] "1742" "177" "572" "867" "1227" "1897" "2662" "318" "1108" "1383"
## [21] "1933" "23" "573" "868" "873" "1228" "1898" "2518" "319" "229"
## [31] "869" "1604" "320" "65" "230" "575" "870" "1900" "3230"
```

#If there are too many outliers, I prefer to only plot the means

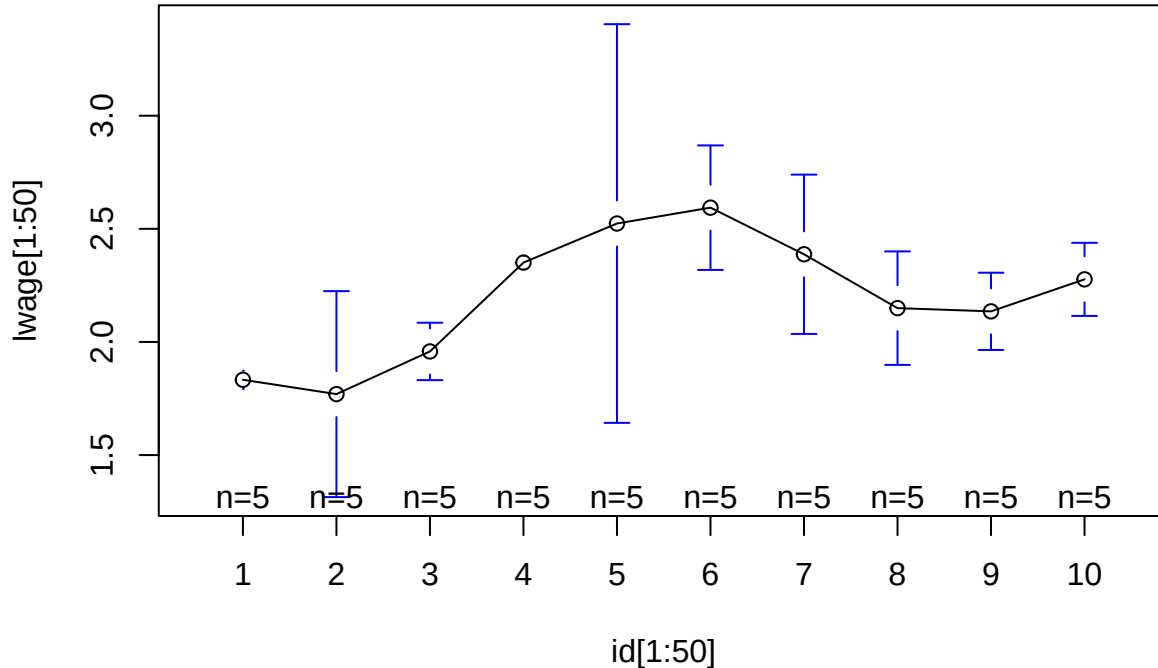
```
plotmeans(lwage ~ year, data = nls.pd)
```



Cleaner result and easy to interpret

Heterogeneity across individuals (look at 10 individuals only)

```
plotmeans(lwage[1:50] ~ id[1:50], data = nls.pd)
```



4 The Fixed Effects Model

- We can extend the model in Eq. 1 to relax the assumption that all individuals have the same coefficients:

$$y_{it} = \beta_{1i} + \beta_{2i}x_{2it} + \beta_{3i}x_{3it} + e_{it} \quad (4)$$

- An i subscript has been added to each of the subscripts, implying that $(\beta_1, \beta_2, \beta_3)$ can be different for each individual
- A popular simplification is one where the intercepts β_{1i} are different for different individuals but the slope coefficients β_2 and β_3 are assumed to be constant for all individuals:

$$y_{it} = \beta_{1i} + \beta_2x_{2it} + \beta_3x_{3it} + e_{it} \quad (5)$$

- All behavioral differences between individuals, referred to as *individual heterogeneity*, are assumed to be captured by the intercept
- Individual intercepts are included to “control” for individual-specific, time-invariant characteristics.
- A model with these features is called a *fixed effects model*
- The intercepts are called *fixed effects*
- We consider two methods for estimating Eq. 5
 - The Least Squares Dummy Variable Estimator
 - The Fixed Effects Estimator

4.1 The Least Squares Dummy Variable Estimator

- One way to estimate the model in Eq. 5 is to include an intercept dummy variable (indicator variable) for each individual

$$D_{1i} = \begin{cases} 1 & i = 1 \\ 0 & \text{otherwise,} \end{cases} \quad D_{2i} = \begin{cases} 1 & i = 2 \\ 0 & \text{otherwise,} \end{cases}, \quad D_{3i} = \begin{cases} 1 & i = 3 \\ 0 & \text{otherwise} \end{cases}$$

- If we have 10 individuals, we define 10 such dummies
- Now we can write:

$$y_{it} = \beta_{11}D_{1i} + \beta_{12}D_{2i} + \cdots + \beta_{1,10}D_{10i} + \beta_2x_{2it} + \beta_3x_{3it} + e_{it} \quad (6)$$

- If the error terms e_{it} are uncorrelated with mean zero and constant variance σ_e^2 for all observations, then the best linear unbiased estimator of Eq. 6 is the least squares estimator
- In a panel data context, it is called the *least squares dummy variable estimator*
- We can test the estimates of the intercepts:

$$\begin{aligned} H_0 &: \beta_{11} = \beta_{12} = \cdots = \beta_{1,10} \\ H_1 &: \text{The } \beta_{1i}\text{'s are not all equal} \end{aligned} \quad (7)$$

- These $N - 1 = 9$ joint null hypotheses are tested using the usual F-test statistic
- In the restricted model all the intercept parameters are equal
- In general, if some $\beta_{1i} \neq \beta_{1j}$ (where $i \neq j$), then there exists some fixed effect, and we can simply do ANOVA (Analysis of Variance) against a pooled model.

Example: Wage Equation (Dummy Variable Approach)

- If we call their common value β_1 , then the restricted model is the pooled model:

$$\begin{aligned} \ln(WAGE) = & \beta_1 + \beta_2EXPER + \beta_3EXPER^2 \\ & + \beta_4TENURE + \beta_5TENURE^2 \\ & + \beta_6UNION + e \end{aligned}$$

where the unrestricted model is:

$$\begin{aligned} \ln(WAGE) = & \beta_{1N} \sum_{N=1}^{10} D_{Ni} + \beta_2EXPER + \beta_3EXPER^2 \\ & + \beta_4TENURE + \beta_5TENURE^2 \\ & + \beta_6UNION + e \end{aligned}$$

- The F-statistic (see R example below for the respective estimates) is:

$$\begin{aligned} F &= \frac{(SSE_R - SSE_U) / J}{SSE_U / (NT - K)} \\ &= \frac{(5.502466 - 2.66719) / 9}{2.66719 / (50 - 15)} \\ &= 4.134 \end{aligned}$$

- The value of the test statistic $F = 4.134$ yields a p -value of 0.0011
- We reject the null hypothesis that the intercept parameters for all individuals are equal.
- We conclude that there are differences in individual intercepts, and that the data should not be pooled into a single model with a common intercept parameter

R Example

```
rm(list=ls())
library(plm) # Fixed Effects (1): Dummy Variable Approach
library(AER)
library(POE5Rdata)

# Read in the data
data("nls_panel", package="POE5Rdata")

# Subset for the first 10 individuals
nls.pd10 <- pdata.frame(nls_panel[nls_panel$id %in% 1:10,])
head(nls.pd10)
```

##		id	year	lwage	hours	age	educ	collgrad	msp	nev_mar	not_smsa	c_city	south
##	1-82	1	82	1.808289	38	30	12	0	1	0	0	1	0
##	1-83	1	83	1.863417	38	31	12	0	1	0	0	1	0
##	1-85	1	85	1.789367	38	33	12	0	0	0	0	1	0
##	1-87	1	87	1.846530	40	35	12	0	0	0	0	1	0
##	1-88	1	88	1.856449	40	37	12	0	0	0	0	1	0
##	2-82	2	82	1.280933	48	36	17	1	1	0	0	0	0
##				black	union	exper	exper2	tenure	tenure2				
##	1-82			1	1	7.666667	58.777777	7.666667	58.777770				
##	1-83			1	1	8.583333	73.67361	8.583333	73.673610				
##	1-85			1	1	10.179490	103.62200	1.833333	3.361111				
##	1-87			1	1	12.179490	148.33990	3.750000	14.062500				
##	1-88			1	1	13.621790	185.55330	5.250000	27.562500				
##	2-82			0	0	7.576923	57.40976	2.416667	5.840278				

```
mreg.fixed10 <- lm(lwage~exper+I(exper^2)+
                    tenure+I(tenure^2)+union+factor(id)-1,
                    data=nls.pd10)
summary(mreg.fixed10)
```

```
##
## Call:
## lm(formula = lwage ~ exper + I(exper^2) + tenure + I(tenure^2) +
##     union + factor(id) - 1, data = nls.pd10)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.95140 -0.07429 -0.02033  0.09319  1.04269
##
```

```
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## exper          0.237997   0.187756   1.268   0.213
## I(exper^2)     -0.008188   0.007905  -1.036   0.307
## tenure        -0.012350   0.034143  -0.362   0.720
## I(tenure^2)     0.002296   0.002688   0.854   0.399
## union          0.113544   0.150863   0.753   0.457
## factor(id)1     0.151917   1.096743   0.139   0.891
## factor(id)2     0.186906   1.071482   0.174   0.863
## factor(id)3    -0.063027   1.350912  -0.047   0.963
## factor(id)4     0.185640   1.343495   0.138   0.891
## factor(id)5     0.938998   1.097777   0.855   0.398
## factor(id)6     0.794496   1.111768   0.715   0.480
## factor(id)7     0.581211   1.235911   0.470   0.641
## factor(id)8     0.537936   1.097495   0.490   0.627
## factor(id)9     0.418345   1.084046   0.386   0.702
## factor(id)10    0.614569   1.090174   0.564   0.577
##
## Residual standard error: 0.2761 on 35 degrees of freedom
## Multiple R-squared:  0.9893, Adjusted R-squared:  0.9847
## F-statistic: 215 on 15 and 35 DF, p-value: < 2.2e-16

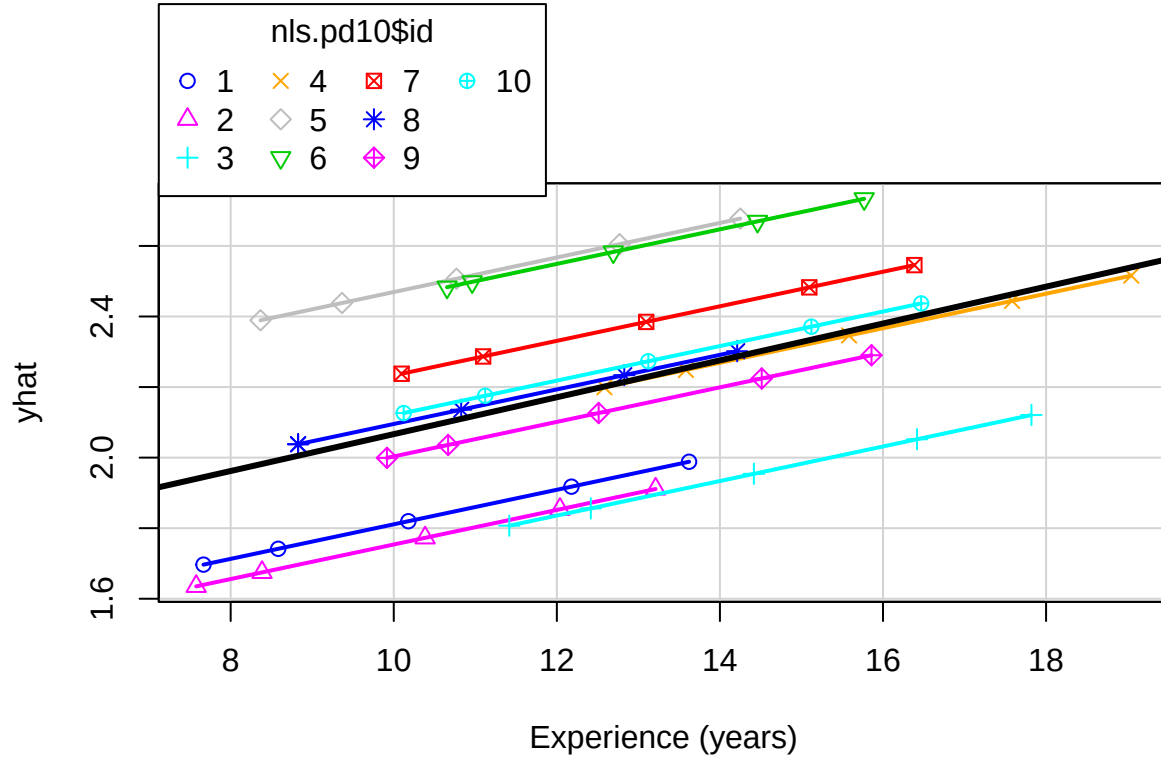
# To see the Dummy Variable effect, we will consider a
# simpler model: lwage ~ education
mregS.fixed10 <- lm(lwage~exper+factor(id)-1,
                    data=nls.pd10)
summary(mregS.fixed10)

##
## Call:
## lm(formula = lwage ~ exper + factor(id) - 1, data = nls.pd10)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.92610 -0.07447  0.01163  0.08221  1.07209
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## exper          0.04897   0.01696   2.888   0.0063 **
## factor(id)1     1.32125   0.21413   6.170 3.00e-07 ***
## factor(id)2     1.26404   0.21235   5.953 6.03e-07 ***
## factor(id)3     1.24805   0.27371   4.560 4.97e-05 ***
## factor(id)4     1.58315   0.29179   5.426 3.25e-06 ***
## factor(id)5     1.97965   0.22345   8.860 7.02e-11 ***
## factor(id)6     1.96141   0.24977   7.853 1.49e-09 ***
## factor(id)7     1.74333   0.25344   6.879 3.15e-08 ***
## factor(id)8     1.60561   0.22345   7.186 1.19e-08 ***
## factor(id)9     1.51359   0.24660   6.138 3.33e-07 ***
## factor(id)10    1.63062   0.25399   6.420 1.35e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2689 on 39 degrees of freedom
## Multiple R-squared:  0.9886, Adjusted R-squared:  0.9854
## F-statistic: 308.7 on 11 and 39 DF, p-value: < 2.2e-16
```

```

yhat <- mregS.fixed10$fitted
scatterplot(yhat ~ nls.pd10$exper | nls.pd10$id,
           xlab = "Experience (years)", ylab = "yhat",
           boxplots = FALSE, smooth = FALSE)
abline(lm(nls.pd10$lwage ~ nls.pd10$exper), lwd=3, col="black")

```



The bold black line is the OLS fit

4.2 The Fixed Effects (Within) Estimator

- Using the dummy variable approach is not feasible when N is large
- Another approach is necessary
- Take the data on individual i :

$$y_{it} = \beta_{1i} + \beta_2 x_{2it} + \beta_3 x_{3it} + e_{it}, \quad t = 1, \dots, T \quad (8)$$

- Average the data across time:

$$\frac{1}{T} \sum_{t=1}^T y_{it} = \frac{1}{T} \sum_{t=1}^T (\beta_{1i} + \beta_2 x_{2it} + \beta_3 x_{3it} + e_{it})$$

- Using the fact that the parameters do not change over time, we can simplify this as:

$$\begin{aligned}
 \bar{y}_i &= \frac{1}{T} \sum_{t=1}^T y_{it} = \beta_{1i} + \beta_2 \frac{1}{T} \sum_{t=1}^T x_{2it} \\
 &\quad + \beta_3 \frac{1}{T} \sum_{t=1}^T x_{3it} + \frac{1}{T} \sum_{t=1}^T e_{it} \\
 &= \beta_{1i} + \beta_2 \bar{x}_{2i} + \beta_3 \bar{x}_{3i} + \bar{e}_i
 \end{aligned} \quad (9)$$

- Now subtract Eq. 9 from Eq. 8, term by term, to obtain:

$$y_{it} - \bar{y}_i = \beta_2 (x_{2it} - \bar{x}_{2i}) + \beta_3 (x_{3it} - \bar{x}_{3i}) + (e_{it} - \bar{e}_i) \quad (10)$$

or

$$\tilde{y}_{it} = \beta_2 \tilde{x}_{2it} + \beta_3 \tilde{x}_{3it} + \tilde{e}_{it} \quad (11)$$

- Usually we are most interested in the coefficients of the explanatory variables and not the individual intercept parameters
- These coefficients can be “recovered” by using the fact that the least squares fitted regression passes through the point of the means
- That is:

$$\bar{y}_i = b_{1i} + b_2 \bar{x}_{2i} + b_3 \bar{x}_{3i}$$

- So that the fixed effects are:

$$b_{1i} = \bar{y} - b_2 \bar{x}_{2i} - b_3 \bar{x}_{3i}, \quad i = 1, \dots, N \quad (12)$$

R Example

```
rm(list=ls())
library(plm) # Fixed Effects (2): Within Approach
library(AER)
library(POE5Rdata)

# Read in the data
data("nls_panel", package="POE5Rdata")

# Subset for the first 10 individuals
nls.pd10 <- pdata.frame(nls_panel[nls_panel$id %in% 1:10,])

# Estimate the within model
mreg.within10 <- plm(lwage~exper+I(exper^2)+
                     tenure+I(tenure^2)+south+union,
                     data=nls.pd10,
                     model="within")
summary(mreg.within10)

## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = lwage ~ exper + I(exper^2) + tenure + I(tenure^2) +
##      south + union, data = nls.pd10, model = "within")
##
## Balanced Panel: n = 10, T = 5, N = 50
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -0.951398 -0.074294 -0.020326  0.093189  1.042695
```

```
##
## Coefficients:
##           Estimate Std. Error t-value Pr(>|t|)
## exper      0.2379965  0.1877561  1.2676  0.2133
## I(exper^2) -0.0081881  0.0079048 -1.0358  0.3074
## tenure     -0.0123500  0.0341433 -0.3617  0.7197
## I(tenure^2) 0.0022961  0.0026885  0.8541  0.3989
## union       0.1135440  0.1508628  0.7526  0.4567
##
## Total Sum of Squares:    3.424
## Residual Sum of Squares: 2.6672
## R-Squared:    0.22103
## Adj. R-Squared: -0.090551
## F-statistic: 1.98628 on 5 and 35 DF, p-value: 0.10501

# We can extract the fixed effects (constants for each individual) using 'fixef'
fixef(mreg.within10)

##           1           2           3           4           5           6           7           8
## 0.151917 0.186906 -0.063027 0.185640 0.938998 0.794496 0.581211 0.537936
##           9          10
## 0.418345 0.614569

# We can now test if we need the fixed effects by performing a
# hypothesis test between the pooled and fixed effects models
# We will use all the observations

nls.pd <- pdata.frame(nls_panel, index=c("id", "year"))
mreg.pooled <- plm(lwage~educ+exper+I(exper^2)+
  tenure+I(tenure^2)+black+south+union,
  model="pooling", data=nls.pd)

mreg.within <- plm(lwage~exper+I(exper^2)+
  tenure+I(tenure^2)+south+union,
  data=nls.pd,
  model="within")

pFtest(mreg.within, mreg.pooled)

##
## F test for individual effects
##
## data:  lwage ~ exper + I(exper^2) + tenure + I(tenure^2) + south + union
## F = 15.188, df1 = 713, df2 = 2858, p-value < 2.2e-16
## alternative hypothesis: significant effects

# Since p-value ~ 0, we reject H0: no effects needed, and
# conclude that the fixed effects are needed
```

5 The Random Effects Model

- In the random effects model we assume that all individual differences are captured by the intercept parameters

- But we also recognize that the individuals in our sample were randomly selected, and thus we treat the individual differences as random rather than fixed, as we did in the fixed-effects dummy variable model
- Random individual differences can be included in our model by specifying the intercept parameters to consist of a fixed part that represents the population average and random individual differences from the population average:

$$\beta_{1i} = \bar{\beta}_1 + u_i \quad (13)$$

- The random individual differences u_i are called *random effects*, and are very similar to the random error term in the OLS regression:

$$E(u_i) = 0, \quad Cov(u_i, u_j) = 0, \quad Var(u_i) = \sigma_u^2 \quad (14)$$

- Substituting Eq. 13 into Eq. 8, we get:

$$\begin{aligned} y_{it} &= \beta_{1i} + \beta_2 x_{2it} + \beta_3 x_{3it} + e_{it} \\ &= (\bar{\beta}_1 + u_i) + \beta_2 x_{2it} + \beta_3 x_{3it} + e_{it} \end{aligned} \quad (15)$$

- Rearranging:

$$\begin{aligned} y_{it} &= \bar{\beta}_1 + \beta_2 x_{2it} + \beta_3 x_{3it} + (u_i + e_{it}) \\ &= \bar{\beta}_1 + \beta_2 x_{2it} + \beta_3 x_{3it} + v_{it} \end{aligned} \quad (16)$$

- The combined error term is:

$$v_{it} = u_i + e_{it} \quad (17)$$

- The random effects error has two components:
 - One for the individual (u_i)
 - One for the regression (e_{it})
- The random effects model is often called an *error components model*
- The combined error term has zero mean:

$$E(v_{it}) = E(u_i + e_{it}) = E(u_i) + E(e_{it}) = 0 + 0 = 0$$

- And a constant, homoskedastic, variance:

$$\begin{aligned} \sigma_v^2 &= Var(v_{it}) = Var(u_i + e_{it}) \\ &= Var(u_i) + Var(e_{it}) + 2Cov(u_i, e_{it}) \\ &= \sigma_u^2 + \sigma_e^2 \end{aligned} \quad (18)$$

5.1 Several Correlations to be Considered:

1. The correlation between two individuals, i and j , at the same point in time, t

$$\begin{aligned} Cov(v_{it}, v_{jt}) &= E[(u_i + e_{it})(u_j + e_{jt})] \\ &= E(u_i u_j) + E(u_i e_{jt}) + E(e_{it} u_j) + E(e_{it} e_{jt}) \\ &= 0 + 0 + 0 + 0 = 0 \end{aligned}$$

2. The correlation between errors on the same individual i at different points in time, t and s

$$\begin{aligned}
Cov(v_{it}, v_{is}) &= E(v_{it}v_{is}) \\
&= E[(u_i + e_{it})(u_i + e_{is})] \\
&= E(u_i^2) + E(u_i e_{is}) + E(u_i e_{it}) + E(e_{it}e_{is}) \\
&= \sigma_u^2 + 0 + 0 + 0 \\
&= \sigma_u^2
\end{aligned} \tag{19}$$

3. The correlation between errors for different individuals in different time periods

$$\begin{aligned}
Cov(v_{it}, v_{js}) &= E(v_{it}, v_{js}) \\
&= E[(u_i + e_{it})(u_j + e_{js})] \\
&= E(u_i u_j) + E(u_i e_{js}) + E(u_j e_{it}) + E(e_{it}e_{js}) \\
&= 0 + 0 + 0 + 0 = 0
\end{aligned}$$

- The errors $v_{it} = u_i + e_{it}$ are correlated over time for a given individual, but are otherwise uncorrelated
- The correlation is caused by the component u_i that is common to all time periods
- It is constant over time and, in contrast to the $AR(1)$ error model, it does not decline as the observations get further apart in time:

$$\begin{aligned}
\rho &= Corr(v_{it}, v_{is}) \\
&= \frac{Cov(v_{it}, v_{is})}{\sqrt{Var(v_{it}) Var(v_{is})}} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_e^2}, \quad t \neq s
\end{aligned} \tag{20}$$

- In terms of the notation introduced to explain the assumptions that motivate the use of cluster-robust standard errors:

$$Var(v_{it}) = \psi_{it} = \sigma_u^2 + \sigma_e^2$$

and

$$Cov(v_{it}, v_{is}) = \psi_{is} = \sigma_u^2, \quad t \neq s$$

- Summary of the error term assumptions of the random effects model:

$$\begin{aligned}
E(v_{it}) &= 0 \\
Var(v_{it}) &= \sigma_u^2 + \sigma_e^2 \\
Cov(v_{it}, v_{is}) &= \sigma_u^2, \quad t \neq s \\
Cov(v_{it}, v_{js}) &= 0, \quad i \neq j \\
Cov(e_{it}, x_{2it}) &= Cov(e_{it}, x_{3it}) = 0 \\
Cov(u_i, x_{2it}) &= Cov(u_i, x_{3it}) = 0
\end{aligned}$$

- We can test for the presence of heterogeneity by testing the null hypothesis $H_0: \sigma_u^2 = 0$ against the alternative hypothesis $H_1: \sigma_u^2 > 0$
- If the null hypothesis is rejected, then we conclude that there are random individual differences among sample members, and that the random effects model is appropriate
- If we fail to reject the null hypothesis, then we have no evidence to conclude that random effects are present
- The Lagrange multiplier (LM) principle for test construction is very convenient in this case
- If the null hypothesis is true, then $u_i = 0$ and the random effects model in Eq. 16 reduces to:

$$y_{it} = \bar{\beta}_1 + \beta_2 x_{2it} + \beta_3 x_{3it} + e_{it}$$

- The test statistic is based on the least squares residuals:

$$\hat{e}_{it} = y_{it} - \bar{b}_1 + b_2 x_{2it} + b_3 x_{3it}$$

- The test statistic for a balanced panels is:

$$LM = \sqrt{\frac{NT}{2(T-1)}} \left[\frac{\sum_{i=1}^N \left(\sum_{t=1}^T \hat{e}_{it} \right)^2}{\sum_{i=1}^N \sum_{t=1}^T \hat{e}_{it}^2} - 1 \right] \quad (21)$$

- If the null hypothesis $H_0: \sigma_u^2 = 0$ is true, then $LM \sim N(0, 1)$ in large samples
- Thus, we reject H_0 at significance level α and accept the alternative $H_1: \sigma_u^2 > 0$ if $LM > z_{(1-\alpha)}$, where $z_{(1-\alpha)}$ is the 100(1 - α) percentile of the standard normal distribution
- This critical value is 1.645 if $\alpha = 0.05$ (since it is a one-tail test)
- Rejecting the null hypothesis leads us to conclude that random effects are present
- We can obtain the generalized least squares estimator in the random effects model by applying least squares to a transformed model:

$$y_{it}^* = \bar{\beta}_1 x_{1it}^* + \beta_2 x_{2it}^* + \beta_3 x_{3it}^* + v_{it}^* \quad (22)$$

where the transformed variables are:

$$\begin{aligned} y_{it}^* &= y_{it} - \alpha \bar{y}_i, & x_{1it}^* &= 1 - \alpha \\ x_{2it}^* &= x_{2it} - \alpha \bar{x}_{2i}, & x_{3it}^* &= x_{3it} - \alpha \bar{x}_{3i} \end{aligned} \quad (23)$$

and α is defined as

$$\alpha = 1 - \frac{\sigma_e}{\sqrt{T\sigma_u^2 + \sigma_e^2}} \quad (24)$$

- α = Fraction of the total variance due to individual heterogeneity
- For $\alpha = 0$, the random effects estimator is identical to the OLS estimator
- For $\alpha = 1$, the random effects estimator is identical to the fixed effects estimator
- For $0 < \hat{\alpha} < 1$, it can be shown that the random effects estimator is a “matrix-weighted average” of the fixed effects estimator that utilizes only within individual variation and a “between estimator” which utilizes variation between individuals
- For example, for our wage equation model, we obtain:

$$\hat{\alpha} = 1 - \frac{\hat{\sigma}_e}{\sqrt{T\hat{\sigma}_u^2 + \hat{\sigma}_e^2}} = 1 - \frac{0.1951}{\sqrt{5 \times 0.1083 + 0.0381}} = 0.7437.$$

6 Comparing Fixed and Random Effects Estimators

- If random effects are present, then the random effects estimator is preferred for several reasons:
 1. The random effects estimator takes into account the random sampling process by which the data were obtained
 2. The random effects estimator permits us to estimate the effects of variables that are individually time-invariant

3. The random effects estimator is a generalized least squares estimation procedure, and the fixed effects estimator is a least squares estimator
- If the random error $v_{it} = u_i + e_{it}$ is correlated with any of the right-hand-side explanatory variables in a random effects model, then the least squares and GLS estimators of the parameters are biased and inconsistent
 - The problem of *endogenous regressors* was considered before
 - The problem is common in random effects models, because the individual specific error component u_i may well be correlated with some of the explanatory variables
 - The panel data regression Eq. 16, including the error component u_i , is:

$$y_{it} = \bar{\beta}_1 + \beta_2 x_{2it} + \beta_3 x_{3it} + (u_i + e_{it}) \quad (25)$$

- Average the observations for each individual over time:

$$\begin{aligned} \bar{y}_i &= \frac{1}{T} \sum_{t=1}^T y_{it} \\ &= \bar{\beta}_1 + \beta_2 \frac{1}{T} \sum_{t=1}^T x_{2it} + \beta_3 \frac{1}{T} \sum_{t=1}^T x_{3it} + \frac{1}{T} \sum_{t=1}^T u_i + \frac{1}{T} \sum_{t=1}^T e_{it} \\ &= \bar{\beta}_1 + \beta_2 \bar{x}_{2i} + \beta_3 \bar{x}_{3i} + u_i + \bar{e}_i \end{aligned} \quad (26)$$

- Subtract Eq. 26 from Eq. 25:

$$y_{it} - \bar{y}_i = \beta_2 (x_{2it} - \bar{x}_{2i}) + \beta_3 (x_{3it} - \bar{x}_{3i}) + (e_{it} - \bar{e}_i) \quad (27)$$

- To check for any correlation between the error component u_i and the regressors in a random effects model, we can use a Hausman test
- The Hausman test can be carried out for specific coefficients, using a t -test, or jointly, using an F -test or a chi-square test
- For the Hausman test, the null hypothesis is that the preferred model is the Random Effects model vs. the alternative that the Fixed Effects model is better.
- Let the parameter of interest be β_k
- Denote the fixed effects estimate as $b_{FE,k}$ and the random effects estimate as $b_{RE,k}$
- The t -statistic for testing that there is no difference between the estimators is:

$$\begin{aligned} t &= \frac{b_{FE,k} - b_{RE,k}}{\sqrt{\widehat{Var}(b_{FE,k}) - \widehat{Var}(b_{RE,k})}} \\ &= \frac{b_{FE,k} - b_{RE,k}}{\sqrt{se(b_{FE,k})^2 - se(b_{RE,k})^2}} \end{aligned} \quad (28)$$

- We expect to find:

$$\widehat{Var}(b_{FE,k}) - \widehat{Var}(b_{RE,k}) > 0$$

- Also:

$$\begin{aligned} Var(b_{FE,k} - b_{RE,k}) &= Var(b_{FE,k}) + Var(b_{RE,k}) \\ &\quad - 2Cov(b_{FE,k}, b_{RE,k}) \\ &= Var(b_{FE,k}) - Var(b_{RE,k}) \end{aligned}$$

- ... because Hausman proved that:

$$\text{Cov}(b_{FE,k}, b_{RE,k}) = \text{Var}(b_{RE,k})$$

- For example, for our wage equation model, applying the t-test to the *SOUTH* we get:

$$\begin{aligned} t &= \frac{b_{FE,k} - b_{RE,k}}{\sqrt{se(b_{FE,k})^2 - se(b_{RE,k})^2}} \\ &= \frac{-0.01632 - (-0.08181)}{\sqrt{(0.03615)^2 - (0.02241)^2}} = 2.31 \end{aligned}$$

- Using the standard 5% large sample critical value of 1.96, we reject the hypothesis that the estimators yield identical results
- Our conclusion is that the random effects estimator is inconsistent, and that we should use the fixed effects estimator, or should attempt to improve the model specification
- The form of the Hausman test in Eq. 28 and its χ^2 equivalent are not valid for cluster robust standard errors, because under these more general assumptions, it is no longer true that:

$$\text{Var}(b_{FE,k} - b_{RE,k}) = \text{Var}(b_{FE,k}) - \text{Var}(b_{RE,k})$$

R Example

```
rm(list=ls())
library(plm)
library(AER)
library(POE5Rdata)

# Read in the data
data("nls_panel", package="POE5Rdata")
nls.pd <- pdata.frame(nls_panel, index=c("id", "year"))

# Fixed Effects Model
mreg.fixed <- plm(lwage~exper+I(exper^2)+
                 tenure+I(tenure^2)+south+union,
                 data=nls.pd,
                 model="within")

# Random Effects Model
mreg.random <- plm(lwage~exper+I(exper^2)+
                  tenure+I(tenure^2)+south+union,
                  data=nls.pd,
                  model="random")

# Test which model is better
phtest(mreg.fixed, mreg.random)
```

```
##
## Hausman Test
##
## data:  lwage ~ exper + I(exper^2) + tenure + I(tenure^2) + south + union
## chisq = 36.218, df = 6, p-value = 2.501e-06
## alternative hypothesis: one model is inconsistent
# Since p-value ~ 0, we reject H0 ("RE is better than FE") and
# --> Choose the Fixed Effects Model instead
```

7 The Hausman-Taylor Estimator

- The Hausman-Taylor estimator is an instrumental variables estimator applied to the random effects model to overcome the problem of inconsistency caused by correlation between the random effects and some of the explanatory variables
- Consider the regression model:

$$y_{it} = \beta_1 + \beta_2 x_{it,exog} + \beta_3 x_{it,endog} + \beta_4 w_{i,exog} + \beta_5 w_{i,endog} + u_i + e_{it} \quad (29)$$

with:

- $x_{it,exog}$ - exogenous variables that vary over time and individual
- $x_{it,endog}$ - endogenous variables that vary over time and individuals
- $w_{i,exog}$ - time-invariant exogenous variables
- $w_{i,endog}$ - time-invariant endogenous variables
- A slightly modify set is applied to the transformed generalized least squares model from Eq. 22:

$$y_{it}^* = \beta_1 + \beta_2 x_{it,exog}^* + \beta_3 x_{it,endog}^* + \beta_4 w_{i,exog}^* + \beta_5 w_{i,endog}^* + v_{it}^* \quad (30)$$

The R example below shows the estimates for a wage equation.

R Example

```
rm(list=ls())
library(plm)
library(AER)
library(POE5Rdata)

# Read in the data
data("nls_panel", package="POE5Rdata")
nls.pd <- pdata.frame(nls_panel, index=c("id", "year"))

# Hausman-Taylor Estimator
# Variables:
# Model: wage equation
# x {it, exog} = exper, exper^2, tenure, tenure^2, union
# x {it, endog} = south
# w {it, exog} = black
# w {it, endog} = educ
# -----
mreg.HT <- plm(lwage~educ+exper+I(exper^2)+
  tenure+I(tenure^2)+black+south+union |
  exper+I(exper^2)+tenure+I(tenure^2)+union+black,
  data=nls.pd, model="ht")
summary(mreg.HT)

## Oneway (individual) effect Hausman-Taylor Model
## (Hausman-Taylor estimator)
##
## Call:
## pht(formula = lwage ~ educ + exper + I(exper^2) + tenure + I(tenure^2) +
##      black + south + union | exper + I(exper^2) + tenure + I(tenure^2) +
##      union + black, data = nls.pd)
##
## T.V. exo  : exper, I(exper^2), tenure, I(tenure^2), union
## T.V. endo  : south
## T.I. exo   : black
## T.I. endo  : educ
##
## Balanced Panel: n = 716, T = 5, N = 3580
##
## Effects:
##               var std.dev share
## idiosyncratic 0.03799 0.19491 0.158
## individual    0.20238 0.44987 0.842
## theta: 0.8098
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -1.1917740 -0.0885848  0.0048605  0.0964688  2.0898352
##
## Coefficients:
##              Estimate Std. Error z-value Pr(>|z|)
## (Intercept) -0.75077101  0.58623628 -1.2807 0.2003121
## educ         0.17050818  0.04446281  3.8348 0.0001256 ***
## exper        0.03990784  0.00647453  6.1638 7.101e-10 ***
```

```
## I(exper^2) -0.00039134 0.00026763 -1.4622 0.1436812
## tenure 0.01432570 0.00315970 4.5339 5.791e-06 ***
## I(tenure^2) -0.00085256 0.00019740 -4.3189 1.568e-05 ***
## black -0.03591345 0.06006813 -0.5979 0.5499209
## south -0.03171219 0.03484738 -0.9100 0.3628064
## union 0.07196924 0.01345447 5.3491 8.839e-08 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares: 772.56
## Residual Sum of Squares: 131.11
## Chisq: 609.256 on 8 DF, p-value: < 2.22e-16

# Note:
# T.V = Time Variant
# T.I = Time Invariant
```

8 Sets of Regression Equations

- Consider procedures for a panel that is long and narrow: T is large relative to N
- If the number of time series observations is sufficiently large, and N is small, we can estimate separate equations for each individual
- These separate equations can be specified as

$$y_{it} = \beta_{1i} + \beta_{2i}x_{2it} + \beta_{3i}x_{3it} + e_{it} \quad (31)$$

- An economic model for describing gross firm investment for the i^{th} firm in the t^{th} time period, denoted INV_{it} , may be expressed as:

$$INV_{it} = f(V_{it}, K_{it}) \quad (32)$$

- We specify the following two equations for General Electric and Westinghouse:

$$\begin{aligned} INV_{GE,t} &= \beta_1 + \beta_2 V_{GE,t} + \beta_3 K_{GE,t} + e_{GE,t} \\ INV_{WE,t} &= \beta_1 + \beta_2 V_{WE,t} + \beta_3 K_{WE,t} + e_{WE,t} \end{aligned} \quad (33)$$

where K = capital stock and V = stock market value of the firm.

- The choice of estimator depends on what assumptions we make about the coefficients and the error terms:
 1. Are the GE coefficients equal to the WE coefficients?
 2. Do the equation errors $e_{GE,t}$ and $e_{WE,t}$ have the same variance?
 3. Are the equation errors $e_{GE,t}$ and $e_{WE,t}$ correlated?
- The assumption that both firms have the same coefficients and the same error variances can be written as:

$$\beta_{1,GE} = \beta_{1,WE}, \quad \beta_{2,GE} = \beta_{2,WE}, \quad \beta_{3,GE} = \beta_{3,WE}, \quad \sigma_{GE}^2 = \sigma_{WE}^2 \quad (34)$$

- Let D_i be an indicator variable equal to one for the Westinghouse observations and zero for the General Electric observations
- Specify a model with slope and intercept indicator variables:

$$\begin{aligned} INV_{it} &= \beta_{1,GE} + \delta_1 D_i + \beta_{2,GE} V_{it} + \delta_2 (D_i \times V_{it}) \\ &\quad + \beta_{3,GE} K_{it} + \delta_3 (D_i \times K_{it}) + e_{WE,t} \end{aligned} \quad (35)$$

- Using the Chow test, we get:

$$\begin{aligned}
 F &= \frac{(SSE_R - SSE_U) / J}{SSE_U / (NT - NK)} \\
 &= \frac{(16,563.0 - 14,989.8) / 3}{14,989.8 / (40 - 6)} \\
 &= 1.189
 \end{aligned}$$

where $NT - NK$ is the total number of degrees of freedom in the unrestricted model

- The p -value for an $F_{(3,34)}$ -distribution is 0.328, implying that the null hypothesis of equal coefficients cannot be rejected
- When both the coefficients and the error variances of the two equations differ, and in the absence of contemporaneous correlation that we introduce in the next section, there is no connection between the two equations, and the best we can do is apply least squares to each equation separately
- Consider the following assumption:

$$Cov(e_{GE,t}, e_{WE,t}) = \sigma_{GE,WE} \neq 0 \quad (36)$$

- The error terms in the two equations, at the same point in time, are correlated
- This kind of correlation is called *contemporaneous correlation*
- The dummy-variable model Eq. 35 represents a way to “stack” the 40 observations for the GE and WE equations into one regression
- To improve the precision of the dummy variable model estimates, we use *seemingly unrelated regressions* (SUR) estimation, which is a generalized least squares estimation procedure
- It estimates the two investment equations jointly, accounting for the fact that the variances of the error terms are different for the two equations and accounting for the contemporaneous correlation between the errors of the GE and WE equations
- There are 3 stages in the SUR estimation procedure:
 1. Estimate the equations separately using OLS
 2. Use the OLS residuals from (1) to estimate σ_{GE}^2 , σ_{WE}^2 and $\sigma_{GE,WE}$

The estimated covariance is given by:

$$\begin{aligned}
 \hat{\sigma}_{GE,WE} &= \frac{1}{\sqrt{T - K_{GE}}\sqrt{T - K_{WE}}} \sum_{t=1}^{20} \hat{e}_{GE,t} \hat{e}_{WE,t} \\
 &= \frac{1}{T - 3} \sum_{t=1}^{20} \hat{e}_{GE,t} \hat{e}_{WE,t} \\
 &= 207.587
 \end{aligned}$$

3. Use the estimates from (2) to estimate the two equations jointly within a generalized least squares framework

Equation	Variable	Coefficient	Std. Error	<i>t</i> -values	<i>p</i> -values
<i>GE</i>	<i>C</i>	−27.7193	29.3212	−0.95	0.351
	<i>V</i>	0.0383	0.0144	2.66	0.012
	<i>K</i>	0.1390	0.0250	5.56	0.000
<i>WE</i>	<i>C</i>	−1.2520	7.5452	−0.17	0.869
	<i>V</i>	0.0576	0.0145	3.96	0.000
	<i>K</i>	0.0640	0.0530	1.21	0.236

Note: *p*-values computed from $t_{(34)}$ distribution.

Figure 1: SUR Estimates of Investment Equations

- The SUR estimation procedure is optimal under the contemporaneous correlation assumption, so no standard error adjustment is necessary
- Two situations in which separate least squares estimation is just as good as the SUR technique
 1. The equation errors are not contemporaneously correlated
If the errors are not contemporaneously correlated, there is nothing linking the two equations, and separate estimation cannot be improved upon
 2. Least squares and SUR give identical estimates when the same explanatory variables appear in each equation
- If the explanatory variables in each equation are different, then a test to see if the correlation between the errors is significantly different from zero is of interest
- Compute the squared correlation:

$$\begin{aligned}
 r_{GE,WE}^2 &= \frac{\hat{\sigma}_{GE,WE}^2}{\hat{\sigma}_{GE}^2 \hat{\sigma}_{WE}^2} \\
 &= \frac{(207.5871)^2}{(777.4463)^2 (104.3079)^2} \\
 &= 0.5314
 \end{aligned}$$

- To check the statistical significance of $r_{GE,WE}^2$, test the null hypothesis $H_0: \sigma_{GE,WE} = 0$
- If $\sigma_{GE,WE} = 0$, then $LM = T \times r_{GE,WE}^2$ is a Lagrange Multiplier test statistic that is distributed as a $\chi^2_{(1)}$ random variable in large samples
- The 5% critical value of a χ^2 -distribution with one degree of freedom is 3.841
- The value of the test statistic is $LM = 10.628$
- We reject the null hypothesis of no correlation
- If we are testing for the existence of correlated errors for more than two equations, the relevant test statistic is equal to T times the sum of squares of all the correlations
- The probability distribution under H_0 is a χ^2 -distribution with degrees of freedom equal to the number of correlations
- With three equations, denoted by subscripts 1, 2 and 3, the null hypothesis is:

$$H_0: \sigma_{12} = \sigma_{13} = \sigma_{23} = 0$$

- The $\chi^2_{(3)}$ test statistic is:

$$LM = T \left(r_{12}^2 + r_{13}^2 + r_{23}^2 \right)$$

- With M equations:

$$LM = T \sum_{i=2}^M \sum_{j=1}^{i-1} r_{ij}^2$$

with $M(M-1)/2$ degrees of freedom

- We previously used the dummy variable model and the Chow test to test whether the two equations had identical coefficients:

$$H_0: \beta_{1,GE} = \beta_{1,WE}, \quad \beta_{2,GE} = \beta_{2,WE}, \quad \beta_{3,GE} = \beta_{3,WE} \quad (37)$$

- It is also possible to test hypotheses such as Eq. 37 when the more general error assumptions of the SUR model are relevant

Because of the complicated nature of the model, the test statistic can no longer be calculated simply as an F -test statistic based on residuals from restricted and unrestricted models

- Most econometric software will perform an F -test and/or a Wald χ^2 -test in a multi-equation framework such as we have here
- In the context of SUR equations both tests are large sample approximate tests
- The equality of coefficients is not the only cross-equation hypothesis that can be tested
- Any restrictions on parameters in different equations can be tested
- Tests for hypotheses involving coefficients within each equation are valid whether done on each equation separately or using the SUR framework
- However, tests involving cross-equation hypotheses need to be carried out within an SUR framework if contemporaneous correlation exists

R Example

```
rm(list=ls())
library(plm)
library(AER)
library(POE5Rdata)
library(systemfit)

# Read in the data and prepare the panel data frame
data("grunfeld2", package="POE5Rdata")
grunf<- grunfeld2
grunf$Firm<-"WE"
for (i in 1:40){
  if(grunf$firm[i]==1){grunf$Firm[i] <- "GE"}
}
grunf$firm <- NULL
```

```

names(grunf)<- c("inv", "val", "cap", "year", "firm")
grunfpd <- plm.data(grunf, c("firm","year"))

# Set of Regression Equations using SUR
grunf.SUR <- systemfit(inv~val+cap, method="SUR", data=grunfpd)
summary(grunf.SUR, resdCov=FALSE, equations=FALSE)

##
## systemfit results
## method: SUR
##
##          N DF      SSR detRCov   OLS-R2 McElroy-R2
## system 40 34 15589.7 35640.6 0.698968   0.615103
##
##      N DF      SSR      MSE      RMSE      R2  Adj R2
## GE 20 17 13788.4 811.081 28.4795 0.692557 0.656388
## WE 20 17  1801.3 105.959 10.2936 0.740401 0.709860
##
## The covariance matrix of the residuals used for estimation
##          GE      WE
## GE 777.446 207.587
## WE 207.587 104.308
##
## The covariance matrix of the residuals
##          GE      WE
## GE 811.081 224.278
## WE 224.278 105.959
##
## The correlations of the residuals
##          GE      WE
## GE 1.000000 0.765043
## WE 0.765043 1.000000
##
##
## Coefficients:
##              Estimate Std. Error t value  Pr(>|t|)
## GE_(Intercept) -27.7193171  29.3212188 -0.94537  0.3577155
## GE_val          0.0383102   0.0144152  2.65763  0.0165755 *
## GE_cap          0.1390363   0.0249856  5.56466 3.4234e-05 ***
## WE_(Intercept) -1.2519882   7.5452174 -0.16593  0.8701684
## WE_val          0.0576298   0.0145463  3.96182  0.0010072 **
## WE_cap          0.0639781   0.0530406  1.20621  0.2442559
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Based on the results, it seems best to estimate the 2 regressions seperately

```